Chapter 5: Periodic Classification of Elements

1. Introduction

At present, 118 elements are known to us. Around the year 1800, only 30 elements were known. All these elements have different properties. To make the study of these elements easy, scientists organized them into groups based on their similarities. This grouping of elements into a systemic table based on common chemical and physical properties is known as the **Periodic Classification of Elements**.

2. Early Attempts at Classification

Dobereiner's Triads

Johann Wolfgang Dobereiner, a German chemist, discovered that elements could be arranged in groups of **three**, called **triads**, which had similar chemical properties.

- **Rule:** When elements are arranged in order of increasing atomic mass, the atomic mass of the middle element is roughly the average of the atomic masses of the other two elements.
- **Example:** Lithium (Li=6.9), Sodium (Na=23), Potassium (K=39). The average of Li and K is roughly 23.
- **Limitations:** He could only identify three triads. This system wasn't applicable to all known elements.

Newlands' Law of Octaves

John Newlands arranged the then-known 56 elements in the order of increasing atomic masses.

- **Law of Octaves:** He found that every eighth element had properties similar to that of the first. He compared this to the octaves found in music.
- **Example:** The properties of Lithium (Li) and Sodium (Na) were found to be the same.

- **Limitations:** It was applicable only up to Calcium. Also, he assumed only 56 elements existed in nature and left no space for future discoveries.

3. Mendeleev’s Periodic Table

Dmitri Ivanovich Mendeleev is credited as the primary developer of the periodic table. He arranged elements on the basis of their fundamental property, the atomic mass, and similarity of chemical properties.

> **Mendeleev's Periodic Law:** The properties of elements are the periodic function of their atomic masses.

- **Features:** The table consists of vertical columns called **groups** and horizontal rows called **periods**.
- **Achievements:**
  1. He completely predicted the existence of elements that had not been discovered at the time (e.g., Scandium, Gallium, Germanium) and left **gaps** for them.
  2. Noble gases could be placed in a new group without disturbing the existing order when they were discovered.
- **Limitations:**
  1. Position of Hydrogen was not fixed (resembled both alkali metals and halogens).
  2. Isotopes violated Mendeleev's law because they have the same chemical properties but different atomic masses.
  3. Some elements with higher atomic mass were placed before those with lower atomic mass (e.g., Cobalt and Nickel).

4. The Modern Periodic Table

In 1913, Henry Moseley showed that the atomic number is a more fundamental property of an element than its atomic mass.

> **Modern Periodic Law:** Properties of elements are a periodic function of their atomic

> number.

- **Structure:** Contains **18 vertical columns (groups)** and **7 horizontal rows (periods)**.
- Elements in a group have the same number of valence electrons and thus similar chemical properties.
- Elements in a period have the same number of electron shells.

Trends in the Modern Periodic Table

- **Valency:** Increases from 1 to 4 and then decreases to 0 across a period. Remains exactly the same down a group.
- **Atomic Size (Radius):** Decreases across a period (due to increased nuclear charge pulling electrons closer) and increases down a group (due to the addition of new shells).
- **Metallic Character:** Decreases across a period and increases down a group. Metals are on the left side, non-metals on the right.
- **Electronegativity:** Increases across a period and decreases down a group.

5. Key Points / Important Notes

- **Metalloids:** Elements that show properties of both metals and non-metals (e.g., Boron, Silicon, Germanium). Located along the zig-zag line.
- **Noble Gases:** Group 18 elements. They have completely filled outermost shells and are highly unreactive.
- **Halogens:** Group 17 elements. Highly reactive non-metals showing a valency of 1.
- The maximum number of electrons that can be accommodated in a shell depends on the formula **$2n^2$** (where n is the shell number).

6. Frequently Asked Questions

Q. Why do elements in the same group have similar chemical properties?

Ans. Because they have the same number of valence electrons in their outermost shell, which determines chemical behavior.

Q. Why does atomic size decrease across a period?

Ans. As we move left to right, the nuclear charge increases. This pulls the electrons tighter toward the nucleus, shrinking the size of the atom.

7. Multiple Choice Questions (MCQs)

1. Upto which element was the Law of Octaves found to be applicable?

a. Oxygen

b. Calcium

c. Cobalt

d. Potassium

Answer: b) Calcium

2. In Mendeleev's Periodic Table, gaps were left for the elements to be discovered later. Which of the following elements found a place later?

a. Germanium

b. Chlorine

c. Oxygen

d. Silicon

Answer: a) Germanium

3. What is the fundamental basis of the Modern Periodic Table?

a. Atomic Mass

b. Number of Nucleons

c. Atomic Number

d. Number of Isotopes

Answer: c) Atomic Number

4. Which of the following statements about the Modern Periodic Table is correct?

a. It has 18 horizontal rows known as periods.

b. It has 7 vertical columns known as periods.

c. It has 18 vertical columns known as groups.

d. It has 7 horizontal rows known as groups.

Answer: c) It has 18 vertical columns known as groups.

5. Which given element A, B, C, D and E with atomic number 2, 3, 7, 10 and 30 respectively belong to the same period?

a. A, B, C

b. B, C, D

c. A, D, E

d. B, D, E

Answer: b) B, C, D

6. Where would you locate the element with electronic configuration 2, 8 in the Modern Periodic Table?

a. Group 8

b. Group 2

c. Group 18

d. Group 10

Answer: c) Group 18

7. An element which is an essential constituent of all organic compounds belongs to:

a. Group 1

b. Group 14

c. Group 15

d. Group 16

Answer: b) Group 14 (Carbon)

8. Which of the following is the outermost shell for elements of period 2?

a. K shell

b. L shell

c. M shell

d. N shell

Answer: b) L shell

9. Which one of the following elements exhibit maximum number of valence electrons?

a. Na

b. Al

c. Si

d. P

Answer: d) P (Phosphorus - 5 valence electrons)

10. Which of the following elements will form an acidic oxide?

a. An element with atomic number 7

b. An element with atomic number 3

c. An element with atomic number 12

d. An element with atomic number 19

Answer: a) An element with atomic number 7 (Non-metals form acidic oxides)

11. The element with atomic number 14 is hard and forms acidic oxide and a covalent halide. To which of the following categories does the element belong?

a. Metal

b. Metalloid

c. Non-metal

d. Left-hand side element

Answer: b) Metalloid (Silicon)

12. Which one of the following does not increase while moving down the group of the periodic table?

a. Atomic radius

b. Metallic character

c. Valence electrons

d. Number of shells in an element

Answer: c) Valence electrons

13. Which of the following property decreases from left to right across the periodic table?

a. Electron affinity

b. Electronegativity

c. Atomic radius

d. Ionization energy

Answer: c) Atomic radius

14. What happens to the electropositive character of elements on moving from left to right in a periodic table?

a. Increase

b. Decreases

c. First increases than decreases

d. First decreases than increases

Answer: b) Decreases

15. The electronic configuration of an element is 2, 8, 4. In modern periodic table, the element is placed in

a. 2nd group

b. 4th group

c. 14th group

d. 18th group

Answer: c) 14th group

16. The atomic mass of the middle element is roughly the average of the atomic masses of the other two elements in:

a. Newland's Octaves

b. Mendeleev's Table

c. Dobereiner's Triads

d. Modern Periodic Table

Answer: c) Dobereiner's Triads

17. Which group elements are called alkali metals?

a. Group 1

b. Group 2

c. Group 17

d. Group 18

Answer: a) Group 1

18. Elements belonging to the same period have the same:

a. Number of valence electrons

b. Number of shells

c. Atomic number

d. Physical properties

Answer: b) Number of shells

19. How many elements are present in the second period of the modern periodic table?

a. 2

b. 8

c. 18

d. 32

Answer: b) 8

20. Which of the following is the most reactive halogen?

a. Fluorine

b. Chlorine

c. Bromine

d. Iodine

Answer: a) Fluorine

8. Short Answer Questions

1. State Modern Periodic Law.

Answer: The Modern Periodic Law states that the physical and chemical properties of the elements are periodic functions of their atomic numbers.

2. How were the positions of isotopes of various elements decided in the Modern Periodic Table?

Answer: Since the Modern Periodic Table is based on atomic numbers, and isotopes of an element have the same atomic number, all isotopes of an element are placed in the same position (same slot) in the Modern Periodic Table.

3. What is a metalloid? Name any two metalloids.

Answer: Elements that exhibit properties of both metals and non-metals are called metalloids. Examples: Silicon (Si) and Germanium (Ge).

9. Long Answer Questions

1. Explain Mendeleev's periodic law. What were the anomalies of Mendeleev's periodic table? How were they resolved in the Modern Periodic Table?

Answer: Mendeleev's Periodic Law states that the properties of elements are a periodic function of their atomic masses.

Anomalies:
1. *Position of Isotopes:* Isotopes have different atomic masses but were placed in the same slot.
2. *Incorrect Order of Atomic Mass:* Elements like Cobalt (58.9) were placed before Nickel (58.7) to match chemical properties.
3. *Position of Hydrogen:* Hydrogen could not be assigned a definite position.

Resolution: The Modern Periodic Table arranged elements by increasing **Atomic Number** instead of mass. This instantly resolved the isotope problem (all isotopes have the same atomic number). Cobalt (27) correctly sits before Nickel (28). Hydrogen's position remains unique but is logically placed at the top of Group 1 due to its 1 valence electron.

2. How does the electronic configuration of an atom relate to its position in the Modern Periodic Table? Explain with an example.

Answer: The electronic configuration gives the exact placement of an element.
- The **number of electron shells** indicates the *Period* number.
- The **number of valence electrons** indicates the *Group* number (for groups 1-2, it's the number; for groups 13-18, it's 10 + number of valence electrons).

Example: Sodium (Na) has an atomic number of 11. Its configuration is 2, 8, 1. It has 3 shells, so it belongs to the 3rd period. It has 1 valence electron, so it belongs to Group 1.